

Customer Shopping Behavior Analysis

Uncovering insights into spending patterns, customer segments, product preferences, and subscription behavior — using transactional data from **3,900 purchases** across multiple product categories to guide strategic business decisions.

DATA ANALYSIS

PYTHON · SQL · POWER BI



Dataset at a Glance

The dataset contains **3,900 rows** and **18 columns**, covering customer demographics, purchase details, and shopping behavior. Only 37 values were missing — all in the *Review Rating* column.

3,900

Total Purchases

Transactional records analyzed

18

Data Columns

Features across demographics & behavior

37

Missing Values

All in Review Rating, imputed by category median

\$50.76

Avg. Purchase

Mean purchase amount (USD)

Customer Demographics

- Age (18–70, mean 44)
- Gender (68% Male, 32% Female)
- Location (50 regions)
- Subscription Status

Purchase & Behavior

- Item, Category, Size, Color, Season
- Purchase Amount (\$20–\$100)
- Discount Applied, Shipping Type
- Previous Purchases, Frequency, Review Rating

Data Preparation in Python

The analysis began with thorough data preparation and cleaning in Python before loading into PostgreSQL for structured querying.

1

Data Loading & Exploration

Imported the dataset using `pandas`. Used `df.info()` to check structure and `.describe()` for summary statistics across all 18 columns.

2

Missing Data Handling

Checked for null values and imputed missing *Review Rating* values using the **median rating of each product category**, preserving category-level rating patterns.

3

Feature Engineering

Created `age_group` by binning customer ages and `purchase_frequency_days` from purchase data. Standardized column names to **snake_case** for consistency.

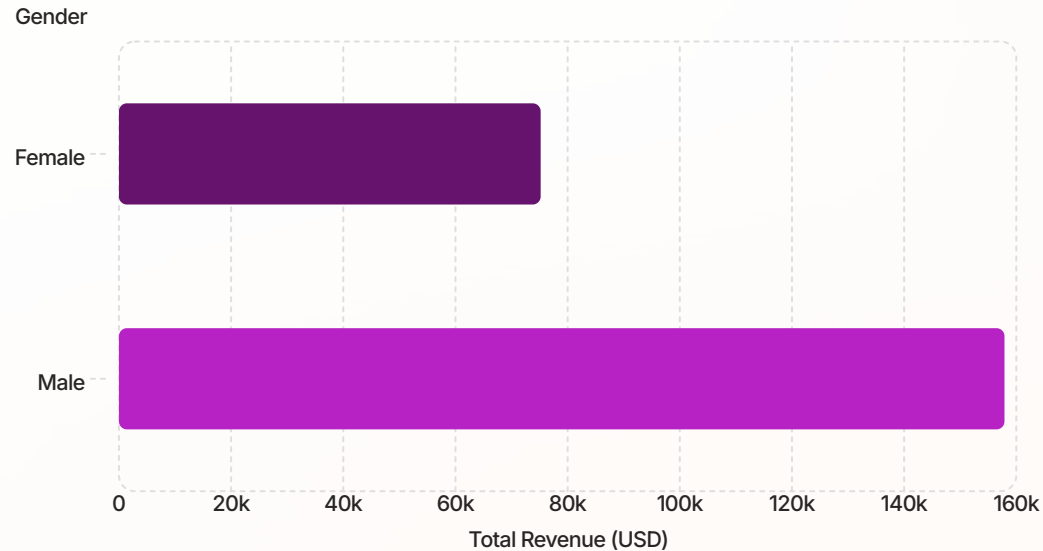
4

Database Integration

Verified `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`. Connected Python to **SQL** and loaded the cleaned DataFrame for SQL analysis.

Revenue by Gender & Discount Users

Revenue by Gender



Male customers generated **\$157,890** in revenue — more than double the **\$75,191** from female customers — despite females representing a meaningful share of the customer base.

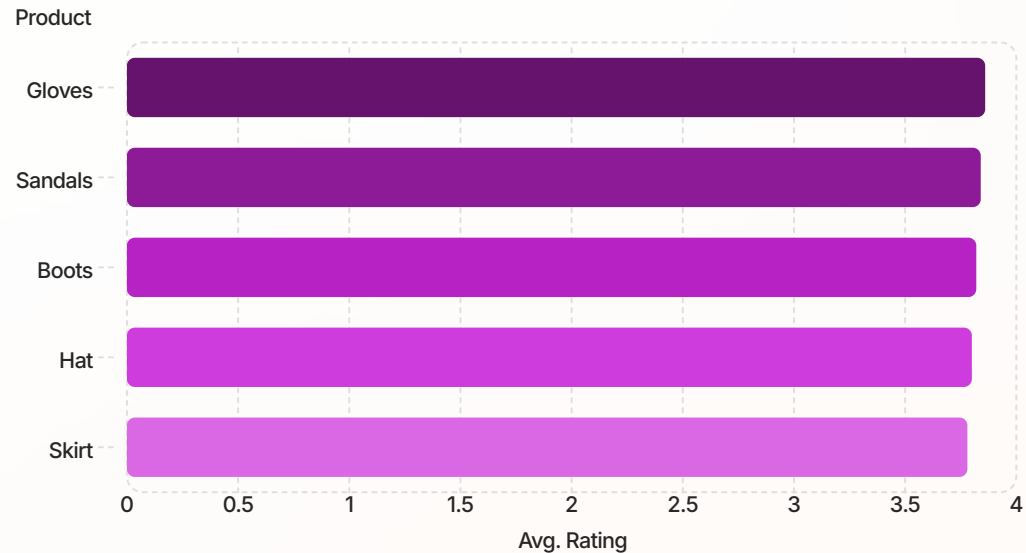
High-Spending Discount Users

839 customers used discounts yet spent **above the average purchase amount**. Top spenders among this group ranged from \$62 to \$99 per transaction.

Key Insight: Discount usage does not necessarily indicate low-value customers. A significant portion of discount users are high spenders, suggesting promotions may drive volume without eroding revenue quality.

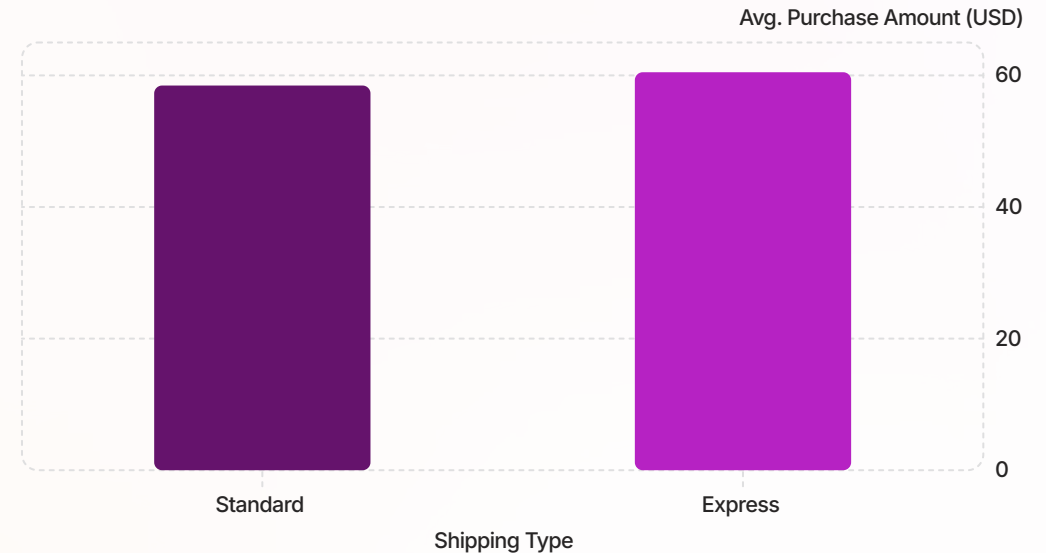
Product Ratings & Shipping Behavior

Top 5 Products by Average Rating



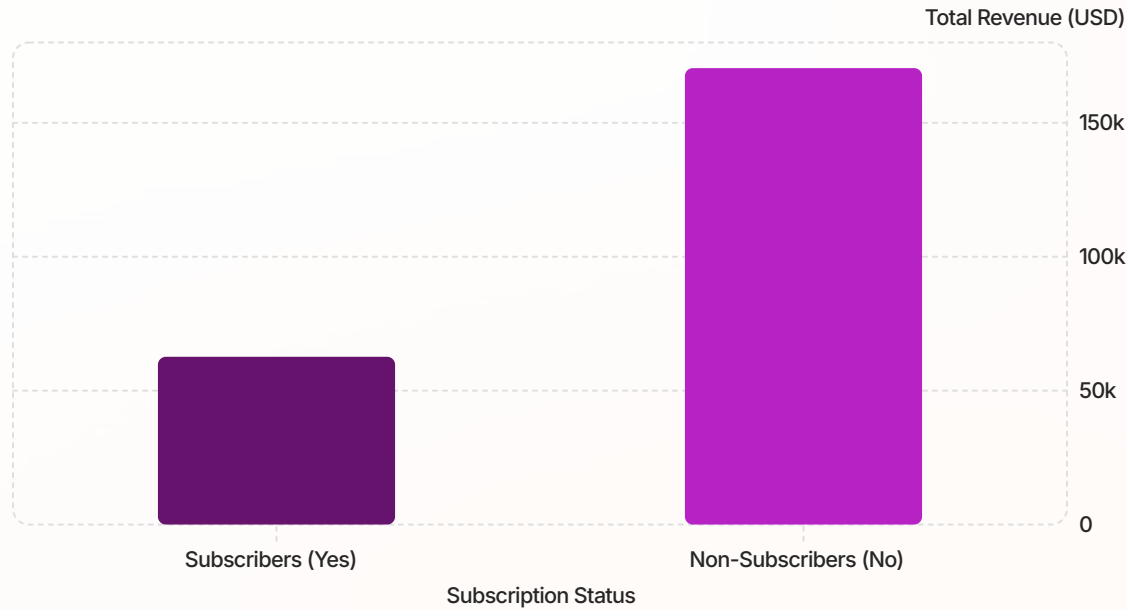
Ratings are tightly clustered between **3.78 and 3.86**, indicating consistent product quality across categories. Overall mean rating is **3.75**.

Shipping Type & Average Spend



Express shipping customers spend slightly more on average (**\$60.48** vs. **\$58.46**), suggesting higher-intent or urgency-driven buyers may represent a premium segment worth targeting.

Subscribers vs. Non-Subscribers



Key Comparison

Subscribers

1,053 customers · Avg. spend: **\$59.49** · Total revenue: **\$62,645**

Non-Subscribers

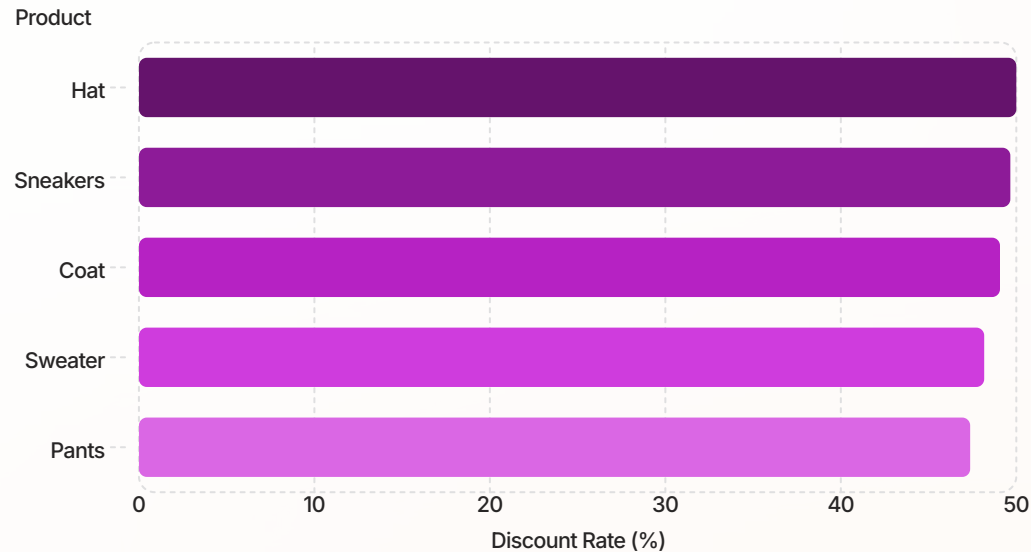
2,847 customers · Avg. spend: **\$59.87** · Total revenue: **\$170,436**



Opportunity: Non-subscribers represent **73% of the customer base** and generate the majority of revenue. Converting even a fraction could significantly boost recurring revenue.

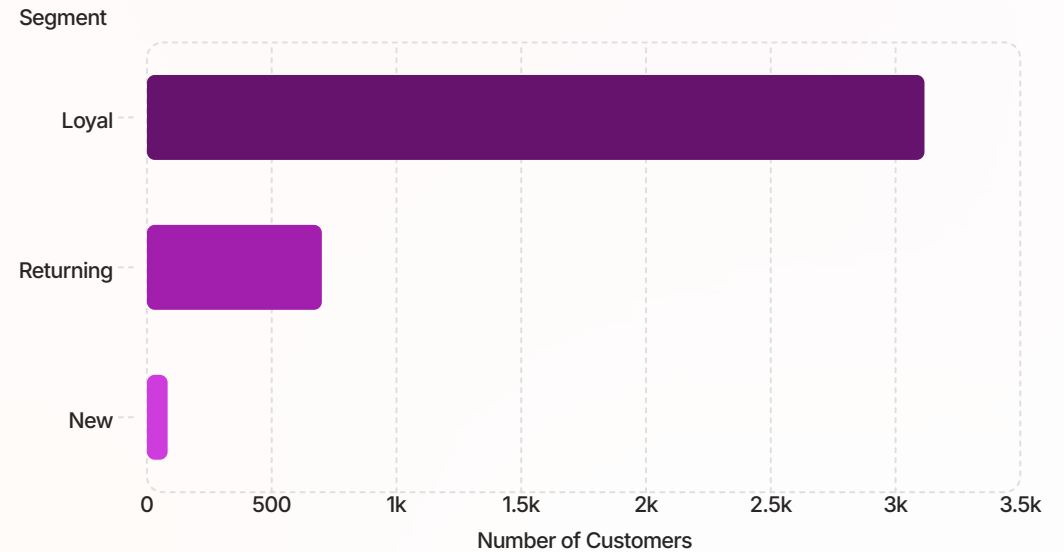
Discount-Dependent Products & Customer Segmentation

Top 5 Products by Discount Rate



These five products have the highest percentage of discounted purchases — all near **50%**. This suggests heavy reliance on promotions, which may impact margins and should be reviewed.

Customer Segmentation



80% of customers are classified as Loyal (3,116), with 701 Returning and only 83 New customers — indicating strong retention but limited new customer acquisition.

Top Products by Category & Repeat Buyers

Top 3 Products per Category

Rank	Category	Product	Orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

Repeat Buyers & Subscriptions

Among customers with **more than 5 purchases**:

- **2,518** are non-subscribers
- **958** are subscribers

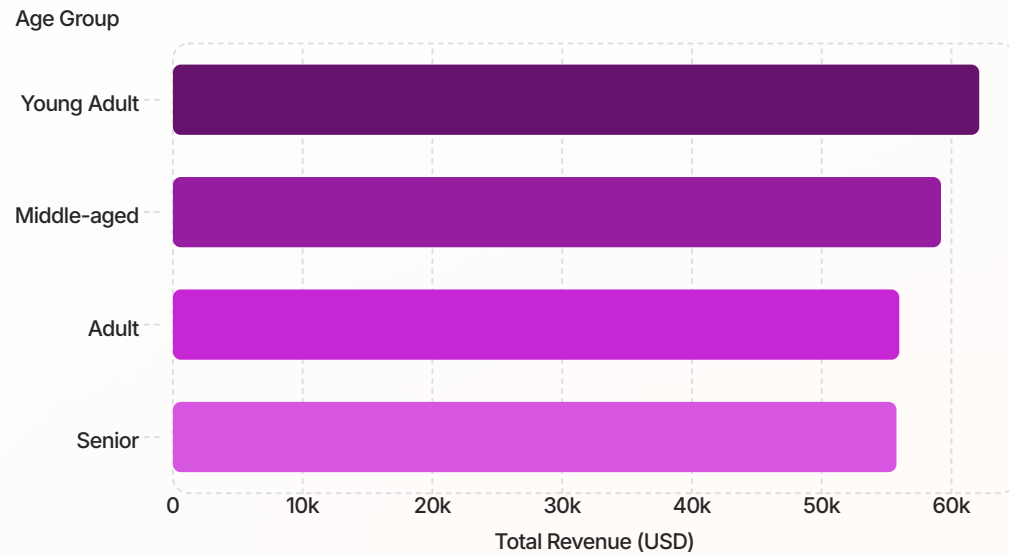
Repeat buyers who are not subscribed represent a prime conversion opportunity — they already demonstrate loyalty but haven't committed to a subscription plan.

Key Takeaway

Blouse, Pants, and Jewelry lead their categories with 171 orders each. Targeting these top performers in campaigns can maximize conversion and revenue impact.

Revenue by Age Group & Power BI Dashboard

Revenue by Age Group



Young Adults lead in revenue contribution at **\$62,143**, though all four age groups are relatively close — suggesting broad appeal across demographics.

Power BI Dashboard Highlights

3.9K Customers

Total customer count

\$59.76 Avg.

Average purchase amount

3.75 Rating

Average review rating

The interactive dashboard includes filters for Subscription Status, Gender, Category, and Shipping Type, with visualizations for revenue by category, sales by age group, and subscription distribution (27% Yes, 73% No).

Business Recommendations



Boost Subscriptions

Promote exclusive benefits for subscribers to convert the 73% non-subscriber base, especially the 2,518 repeat buyers currently without a subscription.



Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment. With only 83 new customers, retention and upsell strategies are critical.



Review Discount Policy

Five products have ~50% discount rates. Balance sales boosts with margin control to ensure promotions are driving profitable growth.



Targeted Marketing

Focus efforts on high-revenue age groups (Young Adults lead at \$62K) and express-shipping users who spend more on average (\$60.48).



Product Positioning

Highlight top-rated products (Gloves 3.86★, Sandals 3.84★) and best-sellers (Blouse, Pants, Jewelry) in marketing campaigns to maximize conversion.